

Characteristic Description

Ordered :Items maintain insertion order Mutable :Elements can be added, removed, or modified Allows Duplicates :Same value can appear multiple times Index-based :Each item has a position starting from index 0 Heterogeneous :Can store different data types together

list

Quick Notes list

- ✓ `append()` → Adds one item
- ✓ `extend()` → Adds multiple items
- ✓ `insert()` → Adds item at specific index
- ✓ `remove()` → Removes by value
- ✓ `pop()` → Removes by index
- ✓ `clear()` → Removes all items
- ✓ `sort()` → Sorts original list
- ✓ `sorted()` → Returns a new sorted list
- ✓ `len()` → Returns list length
- ✓ `sum()` → Adds numeric values

```
In [25]: l=[]  
        l1=[1,2,3,4]
```

```
In [26]: l1
```

```
Out[26]: [1, 2, 3, 4]
```

```
In [27]: l
```

```
Out[27]: []
```

```
In [28]: a = [1, 'c', 2, 's', 3, 's']  
        a
```

```
Out[28]: [1, 'c', 2, 's', 3, 's']
```

```
In [29]: a = [1, 'c', 2, 's', 3, 4, 5, 's']  
        a
```

```
Out[29]: [1, 'c', 2, 's', 3, 4, 5, 's']
```

```
In [30]: a=[1, ['c'], [2, 's', 3], 'm', 4, 's', 6, 's']  
a
```

```
Out[30]: [1, ['c'], [2, 's', 3], 'm', 4, 's', 6, 's']
```

```
In [31]: l2=[1,2,3,4,5,6,['a','b','c'],1+2j,23.2,True]  
l2
```

```
Out[31]: [1, 2, 3, 4, 5, 6, ['a', 'b', 'c'], (1+2j), 23.2, True]
```

```
In [32]: l3=l2.copy()  
l3
```

```
Out[32]: [1, 2, 3, 4, 5, 6, ['a', 'b', 'c'], (1+2j), 23.2, True]
```

```
In [33]: l=['pawan','gaytri',12,34,45,67]  
l
```

```
Out[33]: ['pawan', 'gaytri', 12, 34, 45, 67]
```

```
In [34]: o=[3,4,5,6]  
o
```

```
Out[34]: [3, 4, 5, 6]
```

```
In [35]: print(l2)  
[1, 2, 3, 4, 5, 6, ['a', 'b', 'c'], (1+2j), 23.2, True]
```

```
In [36]: print(id(l2))  
2594084353024
```

```
In [37]: l3
```

```
Out[37]: [1, 2, 3, 4, 5, 6, ['a', 'b', 'c'], (1+2j), 23.2, True]
```

Append

```
In [41]: l3
```

```
Out[41]: [1, 2, 3, 4, 5, 6, ['a', 'b', 'c'], (1+2j), 23.2, True, 0, 34, 'pawan']
```

```
In [42]: l3.append(0)  
l3
```

```
Out[42]: [1, 2, 3, 4, 5, 6, ['a', 'b', 'c'], (1+2j), 23.2, True, 0, 34, 'pawan', 0]
```

```
In [43]: l3.append(34)  
l3
```

```
Out[43]: [1, 2, 3, 4, 5, 6, ['a', 'b', 'c'], (1+2j), 23.2, True, 0, 34, 'pawan', 0, 34]
```

```
In [46]: 13.append('pawan')
13
```

```
Out[46]: [1,
2,
3,
4,
5,
6,
['a', 'b', 'c'],
(1+2j),
23.2,
True,
0,
34,
'pawan',
0,
34,
'pawan',
'pawan',
'pawan']
```

```
In [47]: 14=['pawan','pawan','tanpure','ttt',21,45]
14
```

```
Out[47]: ['pawan', 'pawan', 'tanpure', 'ttt', 21, 45]
```

```
In [60]: ppp=14.append(45,56)
ppp
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[60], line 1
----> 1 ppp=14.append(45,56)
      2 ppp

TypeError: list.append() takes exactly one argument (2 given)
```

clear()

```
In [20]: 14.append('kkk')
14
```

```
Out[20]: ['pawan', 'pawan', 'tanpure', 'ttt', 21, 45, 'kkk']
```

```
In [21]: 14.clear()
14
```

```
Out[21]: []
```

```
In [22]: 14
```

```
Out[22]: []
```

```
In [23]: lrr=[90,909,90,90,90]
lrr
```

```
Out[23]: [90, 909, 90, 90, 90]
```

```
In [24]: lrr.clear()
lrr
```

```
Out[24]: []
```

```
In [54]: y=[1,33,o,'pp']
y
```

```
Out[54]: [1, 33, [3, 4, 5, 6], 'pp']
```

```
In [51]: y=[8,0,0,0,'ooo','jj']
y
```

```
Out[51]: [8, 0, 0, 0, 'ooo', 'jj']
```

```
In [56]: u=y.clear()
u
```

```
In [57]: u
```

```
In [58]: y
```

```
Out[58]: []
```

Extend()

```
In [62]: 14
```

```
Out[62]: ['pawan', 'pawan', 'tanpure', 'ttt', 21, 45]
```

```
In [63]: 14.extend(56,56)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[63], line 1
----> 1 14.extend(56,56)

TypeError: list.extend() takes exactly one argument (2 given)
```

```
In [64]: 14.extend([56,57])
```

In [65]: 14

Out[65]: ['pawan', 'pawan', 'tanpure', 'ttt', 21, 45, 56, 57]

In [76]: eee=14.extend([56,57])
eee

In [77]: 14.extend([56,57])

In [88]: *14

Cell In[88], line 1

*14

^

SyntaxError: can't use starred expression here

In [89]: print(14)

['pawan', 'pawan', 'tanpure', 'ttt', 21, 45, 56, 57, 'o', 56, 'pawan', 56, 'pawan', 56, 57, 58, 56, 57, 56, 57, 56, 57, 56, 57, 56, 57, 56, 57, 1, 2, 3, 1, 2, 3, 90, 90, 90, 0, 90, 0, 90, 90, 90, 90]

In [80]: 15=14.extend([56,57])
15

In [91]: 14.extend([56,56,56])
print(14)

['pawan', 'pawan', 'tanpure', 'ttt', 21, 45, 56, 57, 'o', 56, 'pawan', 56, 'pawan', 56, 57, 58, 56, 57, 56, 57, 56, 57, 56, 57, 56, 57, 56, 57, 56, 57, 1, 2, 3, 1, 2, 3, 90, 90, 90, 0, 90, 0, 90, 90, 90, 90, 56, 56, 56, 56, 56, 56]

In [81]: 15=14.extend([1,2,3])
15

In [82]: 14.extend([1,2,3])

In [87]: print(14)

['pawan', 'pawan', 'tanpure', 'ttt', 21, 45, 56, 57, 'o', 56, 'pawan', 56, 'pawan', 56, 57, 58, 56, 57, 56, 57, 56, 57, 56, 57, 56, 57, 56, 57, 56, 57, 1, 2, 3, 1, 2, 3, 90, 90, 90, 0, 90, 0, 90, 90, 90, 90]

In [84]: 14.extend([90,90,90,0,90,90,90,90,])

In [86]: print(14)

['pawan', 'pawan', 'tanpure', 'ttt', 21, 45, 56, 57, 'o', 56, 'pawan', 56, 'pawan', 56, 57, 58, 56, 57, 56, 57, 56, 57, 56, 57, 56, 57, 56, 57, 56, 57, 1, 2, 3, 1, 2, 3, 90, 90, 90, 0, 90, 0, 90, 90, 90, 90]

index

```
In [92]: l11=[1,2,3,4,5,6,7]
l11
```

```
Out[92]: [1, 2, 3, 4, 5, 6, 7]
```

```
In [95]: l11.index(2)
```

```
Out[95]: 1
```

```
In [96]: l11.index(0)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[96], line 1
----> 1 l11.index(0)

ValueError: 0 is not in list
```

```
In [97]: l11.index(1)
```

```
Out[97]: 0
```

```
In [98]: l11.index(2)
```

```
Out[98]: 1
```

```
In [99]: l11.index(4)
```

```
Out[99]: 3
```

```
In [100... l11.index(5)
```

```
Out[100... 4
```

```
In [101... l11.index(6)
```

```
Out[101... 5
```

```
In [102... l11.index(3,4)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[102], line 1
----> 1 l11.index(3,4)

ValueError: 3 is not in list
```

```
In [103... l11.index(1,2)
```

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[103], line 1  
----> 1 l11.index(1,2)  
  
ValueError: 1 is not in list
```

```
In [104... l11.index(1)
```

```
Out[104... 0
```

```
In [105... l11.index(3)
```

```
Out[105... 2
```

```
In [106... l1=[1,2,3,4,5,6,7]  
l1
```

```
Out[106... [1, 2, 3, 4, 5, 6, 7]
```

Insert()

```
In [107... l1
```

```
Out[107... [1, 2, 3, 4, 5, 6, 7]
```

```
In [108... l1.insert(1,10)
```

```
In [109... l1
```

```
Out[109... [1, 10, 2, 3, 4, 5, 6, 7]
```

```
In [111... l1.insert(0,20)
```

```
In [112... l1
```

```
Out[112... [20, 1, 10, 2, 3, 4, 5, 6, 7]
```

```
In [113... l1.insert(3,33)
```

```
In [114... l1
```

```
Out[114... [20, 1, 10, 33, 2, 3, 4, 5, 6, 7]
```

```
In [115... l1.insert(4,40)
```

```
In [116... l1
```

```
Out[116... [20, 1, 10, 33, 40, 2, 3, 4, 5, 6, 7]
```

```
In [117... l1.insert(5,564)
```

```
In [118... l1
```

```
Out[118... [20, 1, 10, 33, 40, 564, 2, 3, 4, 5, 6, 7]
```

```
In [119... l1.insert(4,44,6,55)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[119], line 1  
----> 1 l1.insert(4,44,6,55)  
  
TypeError: insert expected 2 arguments, got 4
```

```
In [120... l1.insert(4,44:3,30)
```

```
Cell In[120], line 1  
      l1.insert(4,44:3,30)  
                ^  
SyntaxError: invalid syntax
```

```
In [121... l1.insert(5,23(4,44))
```

```
<>:1: SyntaxWarning: 'int' object is not callable; perhaps you missed a comma?  
C:\Users\tanpu\AppData\Local\Temp\ipykernel_8176\590444888.py:1: SyntaxWarning: 'int'  
object is not callable; perhaps you missed a comma?  
      l1.insert(5,23(4,44))  
  
-----  
TypeError                                Traceback (most recent call last)  
Cell In[121], line 1  
----> 1 l1.insert(5,23(4,44))  
  
TypeError: 'int' object is not callable
```

pop

```
In [122... l1
```

```
Out[122... [20, 1, 10, 33, 40, 564, 2, 3, 4, 5, 6, 7]
```

```
In [123... l1.pop()
```

```
Out[123... 7
```

```
In [124... l1.pop()
```

```
Out[124... 6
```

```
In [125... l1.pop()
```


Out[125... 5

In [126... `l1.pop(7)`

Out[126... 3

In [127... `l=[1,2,3,4,5,6,7]`
`l`

Out[127... [1, 2, 3, 4, 5, 6, 7]

In [128... `l.pop(2)`

Out[128... 3

In [129... `l.pop(1)`

Out[129... 2

In [130... `l.pop(0)`

Out[130... 1

In [136... `l`

Out[136... [4, 5, 6, 7]

In [137... `l.pop(1)`

Out[137... 5

In [141... `l23=[10,20,30,40,50,60,70,80]`
`l23`

Out[141... [10, 20, 30, 40, 50, 60, 70, 80]

In [142... `l23.pop(7)`

Out[142... 80

In [143... `l23.pop(5)`

Out[143... 60

In [146... `l23.pop(4)`

Out[146... 50

Remove()

```
In [147... l=[10,20,30,40,50,60,70,80,90]  
l
```

```
Out[147... [10, 20, 30, 40, 50, 60, 70, 80, 90]
```

```
In [148... l.remove(90)
```

```
In [149... l
```

```
Out[149... [10, 20, 30, 40, 50, 60, 70, 80]
```

```
In [150... l.remove(10)
```

```
In [151... l
```

```
Out[151... [20, 30, 40, 50, 60, 70, 80]
```

```
In [152... l.remove(20)
```

```
In [153... l
```

```
Out[153... [30, 40, 50, 60, 70, 80]
```

```
In [154... l.remove(30)
```

```
In [155... l
```

```
Out[155... [40, 50, 60, 70, 80]
```

```
In [156... l.remove(40)
```

```
In [157... l
```

```
Out[157... [50, 60, 70, 80]
```

```
In [158... l.remove(50)
```

```
In [159... l
```

```
Out[159... [60, 70, 80]
```

sort

```
In [161... l20=[100,200,3400,400,5500,50,60,70]  
l20
```

```
Out[161... [100, 200, 3400, 400, 5500, 50, 60, 70]
```

```
In [162... l20.sort()
```

In [163... 120

Out[163... [50, 60, 70, 100, 200, 400, 3400, 5500]

In [164... l=['pawan',23,34,90,12,34,90,1,3,5,'tt']
l

Out[164... ['pawan', 23, 34, 90, 12, 34, 90, 1, 3, 5, 'tt']

In [165... l.sort()

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[165], line 1  
----> 1 l.sort()  
TypeError: '<' not supported between instances of 'int' and 'str'
```

In [166... l=[23,34,90,12,34,90,1,3,5]
l

Out[166... [23, 34, 90, 12, 34, 90, 1, 3, 5]

In [167... l.sort()

In [168... l

Out[168... [1, 3, 5, 12, 23, 34, 34, 90, 90]

In [169... l1=[40,30,50,90,10,20,40,60,30,40]
l1
l1.sort()

In [170... l1

Out[170... [10, 20, 30, 30, 40, 40, 40, 50, 60, 90]

In [171... l1.sort(reverse=True)

In [172... l1

Out[172... [90, 60, 50, 40, 40, 40, 30, 30, 20, 10]

In [173... l23=[10,100,20,30,300,40,500,50,60]
l23
l23.sort()

In [174... l23

Out[174... [10, 20, 30, 40, 50, 60, 100, 300, 500]

In [175... l23.sort(reverse=True)

```
In [176... 123
```

```
Out[176... [500, 300, 100, 60, 50, 40, 30, 20, 10]
```

remove

```
In [177... 123.remove(30)
```

```
In [178... 123
```

```
Out[178... [500, 300, 100, 60, 50, 40, 20, 10]
```

```
In [179... 123.remove(500)
```

```
In [180... 123
```

```
Out[180... [300, 100, 60, 50, 40, 20, 10]
```

```
In [181... 123.remove(10)
```

```
In [182... 123
```

```
Out[182... [300, 100, 60, 50, 40, 20]
```

```
In [183... 123.remove(40)
```

```
In [184... 123
```

```
Out[184... [300, 100, 60, 50, 20]
```

```
In [185... 123.remove(100)
```

```
In [186... 123
```

```
Out[186... [300, 60, 50, 20]
```

indexing and slicing

```
In [187... l10 = [1, 2, 3, 4, 10, 10, 20, 20, 34.5, 34.5, 34.5]
l20 = [1,2,3,4,['a','b','c'],1+3j,34.7,True]
print(l10)
print(l20)
```

```
[1, 2, 3, 4, 10, 10, 20, 20, 34.5, 34.5, 34.5]
[1, 2, 3, 4, ['a', 'b', 'c'], (1+3j), 34.7, True]
```

```
In [188... l10[1]
```

Out[188... 2

In [189... 110[0]

Out[189... 1

In [190... 110[2]

Out[190... 3

In [191... 110[3]

Out[191... 4

In [192... 120[2]

Out[192... 3

In [193... 120[1]

Out[193... 2

In [194... 120[3]

Out[194... 4

In [195... 120[4]

Out[195... ['a', 'b', 'c']

In [196... 120[5]

Out[196... (1+3j)

In [198... 120

Out[198... [1, 2, 3, 4, ['a', 'b', 'c'], (1+3j), 34.7, True]

In [199... 110

Out[199... [1, 2, 3, 4, 10, 10, 20, 20, 34.5, 34.5, 34.5]

In [200... 110[0:6]

Out[200... [1, 2, 3, 4, 10, 10]

In [201... 120[0:6]

Out[201... [1, 2, 3, 4, ['a', 'b', 'c'], (1+3j)]

In [202... 120[3:7]

Out[202...] [4, ['a', 'b', 'c'], (1+3j), 34.7]

In [203...] `l10[3:6]`

Out[203...] [4, 10, 10]

In [204...] `l10[0:7:2]`

Out[204...] [1, 3, 10, 20]

In [205...] `l10[0:8:3]`

Out[205...] [1, 4, 20]

In [206...] `l10[2:8:4]`

Out[206...] [3, 20]

In [207...] `l10[::1]`

Out[207...] [1, 2, 3, 4, 10, 10, 20, 20, 34.5, 34.5, 34.5]

In [208...] `l10`

Out[208...] [1, 2, 3, 4, 10, 10, 20, 20, 34.5, 34.5, 34.5]

In [209...] `l10[::-1]`

Out[209...] [34.5, 34.5, 34.5, 20, 20, 10, 10, 4, 3, 2, 1]

In [210...] `l10[::-3]`

Out[210...] [34.5, 20, 10, 2]

In [211...] `l24=[10,20,30,40,50,60]`
`l24`

Out[211...] [10, 20, 30, 40, 50, 60]

In [212...] `l24[2:-1]`

Out[212...] [30, 40, 50]

In [213...] `l24[2:]`

Out[213...] [30, 40, 50, 60]

In [214...] `l24[1:-2]`

Out[214...] [20, 30, 40]

```
In [215... 124[1]=200  
124
```

```
Out[215... [10, 200, 30, 40, 50, 60]
```

```
In [216... 124[0]=700  
124
```

```
Out[216... [700, 200, 30, 40, 50, 60]
```

```
In [ ]:
```